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Polymer Molecular Weight Analysis by ^{1}H NMR Spectroscopy

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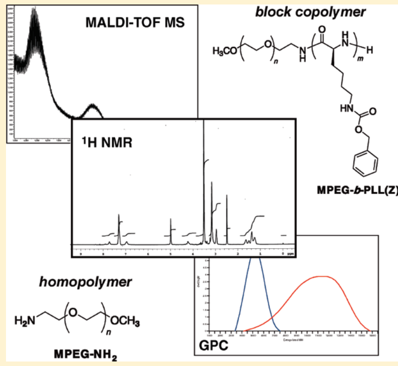
S Supporting Information



ABSTRACT: The measurement and analysis of molecular weight and molecular weight distribution remain matters of fundamental importance for the characterization and physical properties of polymers. Gel permeation chromatography (GPC) is the most routinely used method for the molecular weight determination of polymers whereas matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) is a fast-emerging absolute, and therefore accurate, technique. Although NMR spectroscopy is one of the most powerful tools available for polymer microstructure characterization, among others, its utility in the molecular weight analysis of polymers is somewhat underappreciated. It is a reliable and more readily available teaching tool in comparison to other known techniques, such as GPC and MALDI-TOF MS, for the molecular weight determination of polymers. Demonstrated herein are the simplicity, reproducibility, and convenience of ^{1}H NMR spectroscopy in the analysis of polymer number-average molecular weight (M_n), using α-methoxy-ω-aminopolyethylene glycol (MPEG-NH$_2$) and α-methoxy-polyethylene glycol-*block*-poly-ε-(benzyloxycarbonyl)-L-lysine (MPEG-*b*-PLL(Z)) as model homopolymer and block copolymer, respectively. The molecular weight data from ^{1}H NMR analysis are compared to those from GPC and MALDI-TOF MS.

KEYWORDS: Graduate Education/Research, Upper-Division Undergraduate, Polymer Chemistry, Inquiry-Based/Discovery Learning, Problem Solving/Decision Making, Chromatography, Mass Spectrometry, NMR Spectroscopy

Polymers are large molecules (macromolecules) that are built up of smaller repeating structural units called monomers.[1] They possess an extraordinary range of properties, which aid their essential and ubiquitous roles in virtually all facets of human endeavor. Synthetic polymers are rapidly and successfully replacing many natural polymers because the latter are limited and in short supply, as well as for the former's often-superior physical, chemical, and mechanical properties. Polymers are employed in one form or the other in a wide array of applications such as in food, textiles, electronics, photography, computing, medicine, pharmaceuticals, tissue engineering, and in a variety of biomedical appliances, ranging from implantable devices to controlled drug delivery systems.[2]

Copolymers are frequently formed by the chemical combination of polymers. They benefit from the synergistic characteristics of their constituent (homo)polymers (or monomers) thereby overcoming many deficiencies inherent to their component polymers. Homopolymers are synthesized from single monomers whereas copolymers are prepared from two different monomers (or polymers). Block copolymers, on the other hand, consist of two or more covalently bonded polymer blocks with diverse physicochemical properties. They form phase-separated microdomains in selective solvents and in the bulk.[3] Polyethylene glycol (PEG) is a highly investigated synthetic polymer for the covalent modification of biomacromolecules and surfaces for many pharmaceutical and biotechnical applications,[4] whereas the block copolymers of polyethylene glycol and poly-L-lysine (PEG-*b*-PLL) have been explored for biomedical applications[5] and are considered nontoxic, biocompatible, and biodegradable.[6]

The molecular weight of a polymer represents an average of the distribution of its various constituent molecules with different chain lengths. It is an extremely important variable as it relates directly to the physical properties of the polymer.[7] The determination of molecular weight and molecular weight distribution is therefore of central interest in polymer analysis. Commonly used techniques for the molecular weight determination of polymers include size exclusion chromatography, solution viscosity, osmometry, end-group analysis, ebulliometry, cryoscopy, light scattering, and ultracentrifugation.[8] Comparatively, mass spectrometry has found little use in the polymer field beyond the characterization of degradation products because of its requirement of volatilizable samples.[9] Most polymers thermally degrade before attaining vaporization. Even so, recent years have witnessed the development of new soft ionization systems, such as electrospray ionization (ESI) and matrix-assisted laser desorption/ionization (MALDI) mass spectrometry.[10]

Of the aforesaid, the methods that measure a polymer's colligative properties give the number-average molecular weight (M_n) as the methods count the number of molecules of each weight, whereas the methods that determine molecular weight based on the weight of the constituting molecules give the weight-average molecular weight (M_w). Expressed mathematically,

$$\overline{M} = \frac{\sum N_i M_i^{a+1}}{\sum N_i M_i^{a}} \qquad (1)$$

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where $\overline{M}$ is the average molecular weight; N_i is the number of molecules of species i; and M_i is the molecular weight of species i. From eq 1, $a = 0$ for M_n and $a = 1$ for M_w;[11] that is,

$$M_n = \frac{\sum N_i M_i}{\sum N_i} \quad (2)$$

$$M_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} \quad (3)$$

In other words, M_n is a summation of the product of the mole fraction of each species and its molecular weight, whereas M_w is a summation of the product of the weight fraction of each species and its molecular weight. Thus,

$$M_n = \frac{\sum N_i M_i}{\sum N_i} = \sum X_i M_i \quad (4)$$

where X_i is the mole fraction of species i, and

$$M_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{\sum w_i M_i}{\sum w_i} = \sum W_i M_i \quad (5)$$

where w_i is the weight of species i and W_i is the weight fraction of species i.[12]

Nuclear magnetic resonance (NMR) spectroscopy is a well-established method for the characterization of polymers. It has been applied in the determination of monomer sequence and reactivity ratios and polymer microstructure including stereoregularity, relaxation phenomena, and end-group composition.[13] The role of NMR in polymer composition and microstructure characterization is unrivalled, with its utility in continuous expansion. Conversely, its potential in molecular weight determination is oftentimes underappreciated. Nonetheless, NMR has been applied in the determination of the molecular weight of some polymers.[14]

Gel permeation chromatography (GPC), on the other hand, is the most commonly used technique for the determination of polymer molecular weight.[15] Yet, the underlying principle of separation on the basis of polymer hydrodynamic volume and not molecular weight remains a vital disadvantage in GPC; commensurate monodisperse standards (for calibration) are not always readily available, making the method susceptible to significant errors. Moreover, there are inherent drawbacks with respect to solvent, column, and analysis time; coupled with the fact that it is a relative method, that is, the raw data has to be converted into a molecular weight scale using a calibration procedure obtained under similar conditions to that of the polymer being analyzed. In comparison, ^{1}H NMR is a primary quantitative method, requiring no calibration,[16] and it is routinely run in most laboratories. NMR is therefore a relatively simple, fast, and fairly accurate method of analysis. Also, MALDI mass spectrometry is increasingly becoming a highly accurate and absolute method for macromolecular weight determination but it is fraught with problems, such as mass discrimination, poor reproducibility, and matrix selection.[17]

A particularly useful feature of ^{1}H NMR in molecular weight determination is that the areas under the resonance peaks in the spectra are proportional to the molar concentration of the species in the sample being analyzed.[16] That the area or intensity of the proton signal of a given species is proportional to the amount of that species present in a given sample implies that the area of ith peak is proportional to N_i, the number of molecules of species i with molecular weight M_i, that is,

$$A_i \propto N_i M_i \quad (6)$$

where A_i, N_i, and M_i are the area or intensity of the ^{1}H NMR peak, number of molecules, and molecular weight of species i, respectively. Equation 6 can be rewritten as,

$$A_i = K N_i M_i \quad (7)$$

where K is a constant of proportionality. Assuming that $K = 1$, then $A_i = N_i M_i$ and substituting eqs 2 and 3 become

$$M_n = \frac{\sum A_i}{\sum (A_i / M_i)} \quad (8)$$

$$M_w = \frac{\sum A_i M_i}{\sum A_i} \quad (9)$$

The measurement of a polymer's M_w is dependent on the total weight and total number of polymer particles present in its dilute solution, whereas M_n is dependent on the total number of polymer particles in its dilute solution regardless of polymer size (weight). The determination of M_n is therefore equivalent to dividing the total weight of a given sample of the polymer by the total number of its constituent molecules. So that, mathematically,

$$M_n = \frac{\sum w_i}{\sum N_i} \quad (10)$$

Comparing eqs 5 and 9, $w_i = N_i M_i = A_i$; and substituting for w_i, eq 10 becomes

$$M_n = \frac{\sum A_i}{\sum N_i} \quad (11)$$

where M_n is the number-average molecular weight, A_i is the area or intensity of the ^{1}H NMR peak of species i, and N_i is the number of molecules of species i. Equation 11 establishes the relationship between the number-average molecular weight and the ^{1}H NMR resonance peaks of the detectable hydrogen atoms of a polymer.

It is noteworthy that the determination of M_n by NMR spectroscopy for polymers of molecular weights >25 kDa can be intractable because resolution is diminished the higher the molecular weight. This limitation in measurable M_n is inherent to most colligative methods, and it is due to the loss of sensitivity, at high molecular weights, in the techniques as much as the inability to purify samples and reagents.[18] The development of high-frequency NMR instruments and improved methods of derivatization are, nevertheless, making the measurement of higher M_n possible.[19] Furthermore, the dependence of M_w on a polymer's total mass and number of molecules preclude its measurement by NMR, presently. However, there are auspicious advances in this area.[20]

^{13}C NMR spectra are proton-decoupled and usually better resolved than ^{1}H NMR spectra. They have a large chemical shift (δ) range of 200 ppm but a relatively poor signal-to-noise ratio.[21] M_n can be calculated from ^{13}C NMR spectra providing the NMR experiments are run under quantitative conditions.[16] This can prove cumbersome and requires a little more than a working knowledge of NMR as well as higher sample concentration. In addition, some polymers (e.g., telechelic polymers) cannot be analyzed because many functional group signals (e.g., NH, OH)

Table 1. Salient ^{1}H NMR Data for Molecular Weight Determination of Homopolymer MPEG-NH$_2$ and Block Copolymer MPEG-*b*-PLL(Z)

		MPEG-NH$_2$			MPEG-*b*-PLL(Z)		
Feature	Moiety	δ (ppm)	Multiplicity	Peak area	δ (ppm)	Multiplicity	Peak area
End-groups	CH_3O-	3.25	singlet	0.9	3.26	singlet	0.18
	$-NH_2$	2.71	broad triplet (J = 5.6 Hz)	ND[a]	7.90	singlet	ND
Repeating units	$-(OCH_2CH_2)_n-$	3.52	broad singlet	133.1	3.53	broad singlet	26.71
	$C_6H_5CH_2-$	–	–	–	5.01	singlet	2.00
	ε-CH_2	–	–	–	2.99–2.97	multiplet	2.05
Copolymer linkage	$-CONH-$	–	–	–	7.58	singlet	ND

[a] ND is not determined.

are not detectable in ^{13}C NMR. It is also pertinent to point out that the incidence of poor peak resolution due to the overlap of signals in the ^{1}H NMR spectra do not constitute a significant hindrance to molecular weight determination because of the additive nature of the resonance peaks. However, the accurate integration and assignment of requisite peaks are imperative.

This article highlights the simplicity and convenience of utility of ^{1}H NMR in M_n determination, using α-methoxy-ω-amino-polyethylene glycol (MPEG-NH$_2$) and α-methoxy-polyethylene glycol-*block*-poly-ε-(benzyloxycarbonyl)-L-lysine (MPEG-*b*-PLL(Z)) as model homopolymer and block copolymer, respectively. Requisite working equations are also derived, and the molecular weights of the polymers were determined by size exclusion chromatography (GPC) and mass spectrometry (MALDI-TOF) for comparison. Students can perform the ^{1}H NMR analysis of the two model polymers and the molecular weight calculations in a 3-h laboratory. A student handout is available in the Supporting Information.

COLLECTION AND ANALYSIS OF ^{1}H NMR SPECTRA

A total of 1–5% (w/v) of the polymer, dissolved in deuterated dimethyl sulfoxide (DMSO-d_6), was loaded on a Varian Inova 500 spectrometer (499.8 MHz)[a] at 60 °C to obtain the requisite ^{1}H NMR and (^{1}H–^{1}H) COSY[b] spectra.

The requisite peak areas were obtained by the numerical integration of the ^{1}H NMR spectra, after completing a full assignment of each spectrum and identifying well-resolved peaks.[c] To obtain a reliable baseline for the numerical integration, an isolated resonance signal was chosen as a reference peak, to which a theoretical value was assigned based on the number of protons resonating at that chemical shift (δ). For example, for the block copolymer, MPEG-*b*-PLL(Z), the integration value for the singlet at 5.01 ppm ($C_6H_5CH_2-$; cf. Figure S3 in the Supporting Information) was set to 2.0. The computer-generated peak areas for the protons of the methoxy (CH_3O-) end-group and oxyethylene ($-(OCH_2CH_2)_n-$) repeating units of the homopolymer, MPEG-NH$_2$, were 0.9 and 133.1, respectively. In each case, the values of the integrated peak areas were extracted from the spectrum and divided by their respective numbers of responding protons to afford the relative peak areas. The composition and M_n of the polymers were calculated by substituting the values obtained into the appropriate equations (vide infra).

HAZARDS

Dimethyl sulfoxide is harmful if swallowed, inhaled, or absorbed through skin. It causes irritation to skin, eyes, and respiratory tract. DMSO-d_6 is also hygroscopic and combustible. MPEG-NH$_2$ and MPEG-*b*-PLL(Z) may cause irritation to skin, eyes, and respiratory tract and may be harmful if swallowed or inhaled.

RESULTS FROM ^{1}H NMR SPECTROSCOPY

The degree of polymerization (DP) or number of repeating units of a polymer is determined from its ^{1}H NMR spectrum by comparing the relative proton peak intensity of a known moiety (typically an end-group(s) with a known number of protons) to that of the repeating chain unit of interest. Herein, an equation is derived to calculate the DP as follows: rewriting eq 7,

$$a_i = k n_i m_i \tag{12}$$

and making k the subject of the formula,

$$\frac{a_i}{n_i m_i} = k \tag{13}$$

where a_i is the area or intensity of the ^{1}H NMR peak of species i; n_i is the number of repeating units of species i; m_i is the number of protons of species i; and k is the constant.

Considering the ^{1}H NMR signals of two moieties x and y, from eq 13:

$$\frac{a_x}{n_x m_x} = k_x \tag{14}$$

$$\frac{a_y}{n_y m_y} = k_y \tag{15}$$

where $k_x = k_y$ in a given polymer,

$$\frac{a_x}{n_x m_x} = \frac{a_y}{n_y m_y} \tag{16}$$

where a_x is the area or intensity of the ^{1}H NMR peak of moiety x; n_x is the number of repeating units of moiety x; m_x is the number of protons of moiety x; a_y is the area or intensity of the ^{1}H NMR peak of moiety y; n_y is the number of repeating units of moiety y; and m_y is the number of protons of moiety y. Rearranging eq 16 for n_x,

$$n_x = \frac{a_x m_y n_y}{a_y m_x} \tag{17}$$

where n_x can be used to assess the polymer's DP or number of repeating units.

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Consequently, M_n can be calculated by substituting for n in eq 18,

$$M_n = nM_0 + M_e \tag{18}$$

where n is the number of repeating units or DP; M_0 is the molecular weight of one repeating unit, and M_e is the combined molecular weight of the end-groups.

Some of the data extracted from the ^{1}H NMR spectra of the model polymers for molecular weight determination are collected in Table 1.

^{1}H NMR Spectrum of Homopolymer

The spectrum of MPEG-NH_2 (Figure S2 in the Supporting Information) was well resolved, thereby making its assignment straightforward. The protons of the methoxy (CH_3O-) and amino ($-NH_2$) end-groups resonated as a singlet and broad triplet, respectively, at 3.25 and 2.71 ppm (J = 5.6 Hz), whereas the prominent peak of the oxyethylene protons ($-(OCH_2CH_2)_n-$) was centered at 3.52 ppm.

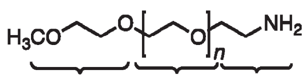
Determination of the Number of Repeating Units

To calculate the number of repeating units (n) in the MPEG-NH_2 chain, the peak areas of CH_3O- (OMe, δ 3.25) and $-(OCH_2CH_2)_n-$ (OEt, δ 3.52) were obtained from the ^{1}H NMR spectrum (cf. Table 1) and appropriately substituted into eq 17:

$$\begin{aligned} n_{\text{MPEG}} &= \frac{a_{\text{OEt}} \cdot m_{\text{OMe}} \cdot n_{\text{OMe}}}{a_{\text{OMe}} \cdot m_{\text{OEt}}} \\ &= \frac{133.1 \times 3 \times 1}{0.9 \times 4} \\ &= 110.92 \approx 111 \end{aligned}$$

Molecular Weight Determination

Having obtained n, the M_n of MPEG-NH_2 was estimated by the summation of the atomic masses of the constituent atoms thus:



Formula Mol. wt. 75.1 44.06n 44.09 g mol^{-1}

$$\begin{aligned} M_n &= 75.1 + 44.06 \times 110.92 + 44.09 \ \text{g mol}^{-1} \\ &= 5006.33 \ \text{g mol}^{-1} \end{aligned}$$

^{1}H NMR Spectrum of Block Copolymer

Distinct signals were observed in the copolymer's spectrum for the protons resonating at 7.58 ppm ($-CONH-$), 7.33–7.32 ppm (C_6H_5-), 6.97 ppm (ε-NH), 5.01 ppm ($C_6H_5CH_2-$), 3.53 ppm ($-(OCH_2CH_2)_n-$), and 3.26 ppm (CH_3O-). The relatively small and partially overlapped singlet for the end-group methoxy protons (CH_3O-) is enlarged (Figure S3 in the Supporting Information).

Determination of the Degree of Polymerization

The determination of the block copolymer's DP was done in two parts: the number of repeating units (n) in the MPEG block was calculated as above, whereupon the degree of polymerization (m) of the PLL(Z) block was obtained. It suffices to compare a well-resolved ^{1}H NMR peak of one of the moieties in the MPEG block ($-(OCH_2CH_2)_n-$ or CH_3O-) to another in the PLL(Z) block ($C_6H_5CH_2-$ or ε-CH_2). Equation 17 afforded the ratio of the proton intensities of CH_3O- (OMe, δ 3.26) and $-(OCH_2CH_2)_n-$ (OEt, δ 3.53):

$$\begin{aligned} n_{\text{MPEG}} &= \frac{a_{\text{OEt}} \cdot m_{\text{OMe}} \cdot n_{\text{OMe}}}{a_{\text{OMe}} \cdot m_{\text{OEt}}} \\ &= \frac{26.71 \times 3 \times 1}{0.18 \times 4} \\ &= 111.29 \approx 111 \end{aligned}$$

The DP of the PLL(Z) block (m) was then similarly calculated by comparing the ^{1}H NMR integrals of $C_6H_5CH_2-$ (Bn, δ 5.01) and $-(OCH_2CH_2)_n-$ (OEt, δ 3.53), using the value of n_{MPEG} calculated above:

$$\begin{aligned} m_{\text{PPL(Z)}} &= \frac{a_{\text{Bn}} \cdot m_{\text{OEt}} \cdot n_{\text{OEt}}}{a_{\text{OEt}} \cdot m_{\text{Bn}}} \\ &= \frac{2.00 \times 4 \times 111.29}{26.71 \times 2} \\ &= 16.67 \approx 17 \end{aligned}$$

Alternatively, comparing ε-CH_2 (εCH_2, δ 2.99–2.97) to CH_3O- (OMe, δ 3.26):

$$\begin{aligned} m_{\text{PPL(Z)}} &= \frac{a_{\varepsilon CH_2} \cdot m_{\text{OMe}} \cdot n_{\text{OMe}}}{a_{\text{OMe}} \cdot m_{\varepsilon CH_2}} \\ &= \frac{2.05 \times 3 \times 1}{0.18 \times 2} \\ &= 17.08 \approx 17 \end{aligned}$$

Determination of Copolymer Composition

The monomer ratio of L-Lys(Z) in the MPEG-*b*-PLL(Z) block copolymer was also estimated from ^{1}H NMR,[7] by comparing ε-CH_2 (from the PLL(Z) block) to $-(OCH_2CH_2)_n-$ (from the MPEG block), using eq 19,

$$\%\text{Lys(Z)} = \frac{\dfrac{a_x}{m_x}}{\dfrac{a_x}{m_x} + \dfrac{a_y}{m_y}} \times 100\% \tag{19}$$

where a_x is the area of the ^{1}H NMR peak of ε-CH_2; m_x is the number of protons of ε-CH_2; a_y is the area of the ^{1}H NMR peak of $-(OCH_2CH_2)_n-$; and m_y is the number of protons of $-(OCH_2CH_2)_n-$. Substituting into eq 19,

$$\begin{aligned} \%\text{Lys(Z)} &= \frac{1.03}{1.03 + 6.68} \times 100\% \\ &= 13.36\% \end{aligned}$$

Comparing $C_6H_5CH_2-$ to $-(OCH_2CH_2)_n-$ similarly, using eq 19, gave a result of 13.02%.

Neither of the peak area values obtained for the end-groups can be substituted into eq 19 to calculate copolymer composition because they are not representative of the polymer. It is equally important to note that eq 19 only holds on the assumption that the relative contributions of the end-groups are negligible. To illustrate, the percentage composition of one of the end-groups (CH_3O-) was estimated to be <1% from its ^{1}H NMR peak by

Table 2. Comparative Molecular Weight Distribution of Homopolymer MPEG-NH_2 and Block Copolymer MPEG-*b*-PLL(Z)

		^{1}H NMR		GPC								MALDI-TOF MS	
		DMSO-d_6[a]		Polystyrene[b]				Polyethylene glycol[b]				Dithranol[c]	α-CHCA[c]
Polymer	Theoretical Molecular Weight	DP	M_n/Da	M_n/Da	M_w/Da	PDI (M_w/M_n)	M_p/Da	M_n/Da	M_w/Da	PDI (M_w/M_n)	M_p/Da	M_p/Da	M_p/Da
MPEG-NH_2	5010 (n = 113)	–	5006	4996	5505	1.102	6158	5030	5122	1.018	5193	4904	4825
MPEG-*b*-PLL(Z)	9470 (n = 113, m = 17)	17	9482	9279	9555	1.030	9822	9686	10220	1.055	11222	–	9987 (4843)[d] (15070)[e]

[a] Solvent. [b] Calibrant. [c] Matrix. [d] $[M_n]^+$. [e] $[M_nM_m]_3^{2+}$.

appropriately substituting into eq 19:

$$\%\mathrm{OMe} = \frac{0.06}{0.06 + 6.68} \times 100\% = 0.89\%$$

To check the accuracy of the previous DP calculation (from eq 17), eq 20 was used,

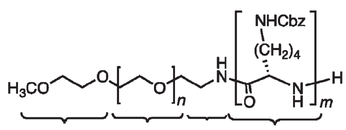
$$\frac{m}{n} = \frac{z}{100\% - z} \qquad (20)$$

where m is the degree of polymerization of the PLL(Z) block; n is the number of repeating units of the MPEG block; and z is the percentage monomer ratio of the PLL(Z) block.

$$m = \frac{111.29 \times 13.36\%}{86.64\%} = 17.16 \approx 17$$

Molecular Weight Determination

With the respective values of n and m of the MPEG and PLL(Z) blocks known, the M_n of the block copolymer, MPEG-*b*-PLL(Z), was assessed:



Mol. wt. Formula: 75.1   44.06n 43.08   262.34m 1.01 g mol^{-1}

$$M_n = 75.1 + 44.06 \times 111.29 + 43.08 + 262.34 \times 17 + 1.01 \text{ g mol}^{-1} = 9482.41 \text{ g mol}^{-1}$$

■ MOLECULAR WEIGHT DETERMINATION BY ALTERNATIVE METHODS

Gel Permeation Chromatography

The GPC experiments to determine the molecular weights of MPEG-NH_2 and MPEG-*b*-PLL(Z) were carried out with BHT-stabilized THF as mobile phase (cf. Figure S4 in the Supporting Information). Twelve polystyrene (PS) and five PEG narrow polydispersity standards were used as calibrants to obtain the universal calibration curves. The PEG calibration curve gave a better polydispersity index (PDI) than the PS curve for MPEG-NH_2. The converse was the case for MPEG-*b*-PLL(Z). The apparently better compatibility exhibited by the PS standards toward MPEG-*b*-PLL(Z) may be attributable to the aryl moiety, which is common to both calibrant and analyte copolymer.

The M_n values obtained from the PS-calibrated curve were lower than the theoretical molecular weights for MPEG-NH_2 and MPEG-*b*-PLL(Z), whereas the opposite scenario was evident with the PEG standards (cf. Table 2). Comparing the experimental and calculated values for M_w, the PS calibrant gave a higher deviation for MPEG-NH_2 but a much lower deviation for MPEG-*b*-PLL(Z). The PEG curve also overestimated the block copolymer's M_w. It is instructive to note that because the constituent blocks in a copolymer usually have different refractive indices, the GPC elution curve is only truly representative of its molecular weight distribution if the chemical composition of the block copolymer is invariant with the elution volume.[22]

MALDI-TOF Mass Spectrometry

The molecular weights of MPEG-NH_2 and MPEG-*b*-PLL(Z) were also measured by MALDI-TOF MS, using two different matrix systems: 1,8-dihydroxyanthrone (dithranol) and α-cyano-4-hydroxycinnamic acid (α-CHCA). Significant successes have been recorded in the application of the dithranol matrix in the MALDI MS of synthetic polymers in the past decade[10] while α-CHCA has found wide utility in the analysis of peptides and proteins.[23]

MALDI-TOF Mass Spectra of Homopolymer

The MS of MPEG-NH_2, obtained with sodium cationization and dithranol, gave an envelope of partially mass-resolved signals centered at 4900 Da (Figure S5 in the Supporting Information). Two clusters of signals (labeled **i** and **ii**) were observed. Similar peak clusters were separated by a nominal value of 44 Da, which is consistent with the $-(OCH_2CH_2)_n-$ repeating unit of MPEG-NH_2. Furthermore, the **i** and **ii** peaks were separated by 16 Da; implying the presence of nonsodiated (**i**) and sodiated (**ii**) polymer chains. The nonsodiated peaks were probably due to adventitious lithium adducts of the polymer.[24] The onset and end of the high signal intensity were observed around 4 and 6 kDa, respectively.

The combined molecular weight of the homopolymer's end-groups was estimated from its mass spectrum,[10,25] as exemplified below, using the peak at 4903.8027 (M_p). To calculate the combined molecular weight of the end-groups (M_e) from the selected m/z signal, the atomic mass of sodium was first subtracted from the mass of the peak, and the result was then divided by the molecular weight of the repeating unit to afford the number of repeating units (n) of the polymer chain:

$$4903.80 - 22.99 = 4880.81$$
$$\frac{4880.81}{44.06} = 110.776$$
$$0.776 \times 44.06 = 34.19$$
$$Check: M_e = M_{CH_3} + M_{NH_2} = 15.04 + 16.03 = 31.07 \text{ Da}$$

The product of the residual value of n (i.e., 0.776) and the molecular weight of the repeating unit gave the combined mass of both end-groups as 34.19 Da, which is close to the total mass of the methyl and amino end-groups. The difference (ca. 3 Da) between the experimental and calculated values is isotope-related.

The MS of MPEG-NH_2 in the α-CHCA matrix was also examined (Figure S6 in the Supporting Information). It proved to be better resolved than the dithranol-based spectrum, showing the polymer chains present as mass-resolved signals. The expanded portion of the spectrum revealed three major series of peaks (**i**, **ii**, and **iii**), which were each 44 Da apart. Within each set of peaks having the same number of repeating units, **i** and **ii** were lower than **iii** by 33 and 20 Da, respectively. The peaks were assigned, based on the n and M_e calculations (vide supra), after an extensive series of permutations of the adventitious cations (such as, Li^+, Na^+, and K^+)[17] with the expected H^+ ions. An underlying factor in the calculations and subsequent assignment of peaks was the proximity of the calculated values to the theoretical M_e (31.07 Da).

Considering the six consecutive peaks in the m/z 4800 region (Table S1 in the Supporting Information), for example, the signals at m/z 4825 and 4869 were assigned to the MPEG-NH_2 polymer chain with 108 and 109 oxyethylene repeating units, respectively, with proton, lithium and sodium adducts ($M^+ + 5H^+ + Li^+ + Na^+$).[26] The peaks at m/z 4836 and 4880 had two proton adducts ($M^+ + 2H^+$); so that n = 109 and 110, respectively. The m/z 4849 and 4893 signals, respectively, corresponded to 109 and 110 oxyethylene repeating units, as well, but were cationized with two adducts of lithium and a proton ($M^+ + H^+ + 2Li^+$). Consequently, the MS signals at m/z 4836, 4849, and 4869 can be attributed to the MPEG-NH_2 chain with 109 repeating units of oxyethylene (i.e., $-(OCH_2CH_2)_{109}-$). The agreement of the experimental and theoretical data lends credence to the validity of this approach. Nevertheless, requisite software is being developed[27] as the foregoing approach is laborious.

MALDI-TOF Mass Spectrum of Block Copolymer

The MALDI-TOF MS of MPEG-*b*-PLL(Z) with the dithranol matrix, and CF_3COONa as cationization agent, neither gave signals in the linear nor reflectron modes. The reason for this has yet to be elucidated, as peptides have been previously analyzed using dithranol.[28] A partially resolved spectrum (Figure S7 in the Supporting Information) was, however, obtained with the α-CHCA matrix. The MS showed three major envelopes of signals centered on 5, 10, and 15 kDa. In the first envelope in the m/z 4−6 kDa region, two series of signals, each bearing a repeating unit of 44 Da, were discernible. The other envelopes of intense signals in the higher mass region were not mass-resolved.

On the basis of the observation of peaks separated by 44 Da, which is consistent with the oxyethylene repeating unit, in the MALDI-TOF MS of MPEG-*b*-PLL(Z), it appears that a main fragmentation process involving the cleavage of the block copolymer's ethylamide linkage ($-OCH_2CH_2NHCO-$) to the two constituent blocks (i.e., MPEG and PLL(Z)) may have occurred. Przybilla et al.[29] reported a similar observation for poly(ethylene oxide)-*b*-poly(*p*-phenylene ethynylene). Notably, the lowering of the baseline observed at high m/z in the MS is more of a reflection of the low molecular weight component of samples than a saturation of the detector.[17]

In Table 2, the nominal molecular weight values of the homopolymer, MPEG-NH_2, and block copolymer, MPEG-*b*-PLL(Z), obtained from ^{1}H NMR, GPC, and MALDI-TOF MS analyses are compared.

■ CONCLUSION

The utility of ^{1}H NMR in M_n determination of polymers has been discussed. Some known equations have been modified and requisite new equations have been developed. The average molecular weights of the model polymers were also determined by GPC and MALDI-TOF MS, and the concomitant data were compared with good agreement. The M_n determinations of MPEG-NH_2 and MPEG-*b*-PLL(Z) highlight the application of ^{1}H NMR in this area. ^{1}H NMR is rapidly becoming routine and remains unrivalled in the elucidation of polymer microstructure. The current state-of-the-art also bodes well for the surmounting, presently, of one of its foremost drawbacks—the inability to measure M_w. ^{1}H NMR spectroscopy is a fast, reproducible, and relatively simple technique.

Although GPC is the most widely used method for the determination of molecular weight distribution of polymers, the strong dependence of its data on the calibrant, as exemplified by the model polymers herein, remains an encumbrance. Additionally, GPC is a relative method, requiring a molecular weight detector, and can be used only in conjunction with other absolute methods. It is equally noteworthy that in block copolymers, for example, where detailed information about segment composition and block length are pertinent,[30] GPC is incapable of measuring the molecular weight of the constituting blocks. The molecular weight analysis of polymers by MALDI-TOF MS, on the other hand, is a relatively new and auspicious (absolute) method but it is plagued by a number of shortcomings, albeit mostly nascent. The difficulties and uncertainties associated with matrix choice, sample preparation, and the interpretation of the mass spectra, for instance, detract from its potential benefits.

It is incontrovertible that each of the methods discussed above has pros as well as cons with respect to the molecular weight determination of polymers, and ideally complement one another. In a hypothetical scenario of mutual exclusivity, however, ^{1}H NMR has the advantage of not only being simple and routine, but fundamentally multifarious, yielding reliable information regarding polymer microstructure and molecular weight, among others. ^{1}H NMR is a more readily available teaching and learning tool in comparison to GPC and MALDI-TOF MS.

■ ASSOCIATED CONTENT

Supporting Information

This includes a student laboratory handout and details of the analytical parameters, calculations, mechanism of polymerization, and spectra (^{1}H NMR, GPC, MALDI-TOF MS) for the model polymers. This material is available via the Internet at http://pubs.acs.org.

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■ ADDITIONAL NOTE

[a] A low-frequency ^{1}H NMR spectrometer can be used for the molecular weight analysis of polymers ($M_n \leq 25{,}000$) providing the salient proton resonance peaks are properly resolved for their accurate assignment and subsequent comparison.

[b] Two-dimensional NMR experiments, such as the proton−proton correlation spectroscopy (COSY), facilitate accurate peak assignments.

[c] The molecular weight determination of more complex polymer systems (e.g., random copolymers) by ^{1}H NMR can be quite challenging because of the polymers' often-complicated spectra. In cases as these, knowledge of the polymer's mechanism of polymerization is a prerequisite (see the Supporting Information).

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